

Ultrafiltration in Food Processing Industry: Review on Application, Membrane Fouling, and Fouling Control

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Abstract Ultrafiltration process has been applied widely in food processing industry for the last 20 years due to its advantages over conventional separation processes such as gentle product treatment, high selectivity, and lower energy consumption. Ultrafiltration becomes an essential part in food technology as a tool for separation and concentration. However, membrane fouling compromises the benefits of ultrafiltration as fouling significantly reduces the performance and hence increases the cost of ultrafiltration. Recent advances in this area show the various intensive studies carried out to improve ultrafiltration, focusing on membrane fouling control and cleaning of fouled membranes. Thus, this paper reviews recent developments in ultrafiltration process, focusing on fouling mechanisms of ultrafiltration membranes as well as the latest techniques used to counter membrane fouling.

Keywords Ultrafiltration · Membrane technology · Fouling control · Food processing industry

Introduction

Membrane filtration processes have gained popularity in the food processing industry over the last 25 years. It is estimated that 20–30% of the current €250 million turnover of membrane used in the manufacturing industry worldwide was from food processing industry. To date, this market is

still undergoing rapid growth, approximately 7.5% per year, particularly in dairy industry, followed by beverages and egg products. The total membrane market for the food and beverages industry has been estimated to be worth US \$1,182 billion in 2008 (Sutherland 2004). In the dairy industry, it is estimated that over 75% of membrane usage is dedicated to whey processing, while 25% of ultrafiltration (UF) membranes is accounted for milk processing (Eykamp 1995; Timmer and Van der Horst 1998).

Compared to conventional competitive concentration (thermal processes) and separation operations (decantation, filtration, centrifugation, chromatography, etc.), membrane separation processes are of great interest and attractive to industry due to three main benefit categories as follows (Daufin et al. 2001; Lim and Mohammad 2011):

- (a) Higher quality of process food—Customer requirements for food have evolved with safety + novelty + diversity + nutrition. This evolution necessitates the design of novel foods and intermediate food products by manufacturing fractions and co-fractions from initial products. Moreover, membrane separation process could preserve the nutrition of fresh food with lower risk of contamination.
- (b) Competitiveness and economical consideration—In preparation of traditional food products, membrane processes contribute to simplification of process flow (reduce some production steps) and improvement of production processes (removes unwanted ingredients like food contaminants that have a negative impact on product quality, making the final product more attractive in texture and increasing its shelf-life) and food quality (mild temperature operation with non-destructive for thermally labile foods and flavors). Moreover, membrane processes are simple, easy to implement, and modular systems in nature (which are compact yet have great flexibility with good automation).

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- (c) Environmentally benign—Membrane processes eliminate the use of polluting materials (diatomaceous earth, DE) in clarification of wine, beer, fruit juices, etc. The use of DE results in a number of problems, including health and environmental concerns regarding dust exposure and issues related to the disposal of spent cake to landfill.

Unfortunately, membrane fouling caused by the deposition of biological suspensions or macromolecules/colloids/particles on the membrane surface or into the membrane pores limits the widespread application of membrane separation in food processing industry. Membrane fouling results in substantial flux decline and increase of plant maintenance and operating costs, including the need for pretreatment, membrane cleaning, limited recoveries and feed water loss, and short lifetime of membranes. Therefore, the objective of this review is to systematically provide an overview of recent development of ultrafiltration in food processing industry and the associated membrane fouling, focusing on the methods to reduce fouling, challenges and development of fouling control methods, and **treatment for flux recovery**. These aspects have not been addressed by any reviews previously especially for those related to food industry. The recent review by Goosen et al. (2004) has been on membrane fouling for desalination application.

Application of Ultrafiltration in Food Industry

The applications of membrane processes in food industry can be classified into three main areas namely dairy industry, beverage industry and fish and poultry industry (Chabeaud et al. 2009; Daufin, et al. 2001; Pouliot 2008). This simple classification highlights the versatility acquired by the membrane processes over the years and their wide range of applications in the food industry.

Dairy Industry

The contemporary use of membranes in dairy processing had been reviewed in International Dairy Federation special issue published in 2004 and by some other authors (Daufin, et al. 2001; Fox et al. 2004; Moresi and Lo Presti 2003; Pouliot 2008; Rosenberg 1995; Saxena et al. 2009). The dairy industry has been one of the pioneers in the development of UF equipment and techniques based on the experience gained from its application in the dairy processing field. UF has found a major application in the production of cheese. Initially, during cheese production, whey was discharged to the sewer due to its high salt and lactose content, causing the direct use as a food supplement difficult, but now whey can be processed to obtain additional food values through a newer process using UF membrane by increasing the fraction of milk proteins used

as cheese or some other useful products and reduce the waste disposal problem represented by whey (Saxena, et al. 2009).

Beverages Industry

Membrane technology is recognized as a standard tool in the food and beverage industry (Cheryan 1998). It is being employed for processing a variety of fruit and vegetable juices (orange, lemon, grapefruit, tangerine, tomato, cucumber, carrot, and mushroom) (Echavarria et al. 2011). In juice clarification, ultrafiltration can be used to separate juices into fibrous concentrated pulp (retentate) and a clarified fraction free of spoilage microorganisms (permeate). The pasteurized clarified fraction can then undergo non-thermal membrane concentration and eventually whole juice reconstitution by combination with pasteurized pulp, in order to obtain a product with improved organoleptic qualities (Cassano et al. 2008). Also, a superior quality clarified fruit juice could make a strong impact in new market areas, such as clear juice blends, liqueur, and related products such as carbonated soft drinks, and in all applications where suspended solids have a negative effect on final product quality (de Barros et al. 2003). Apart from that, ultrafiltration is also applied to the concentration process in fruit juice processing industry. Ultrafiltration has been proved to recover bioactive components in fruit juice. Galaverna et al. (2008) studied the influence of ultrafiltration on the composition of these bioactive compounds in order to develop a natural product, which is used to fortify foods and beverages. They found that most bioactive compounds of the depectinized kiwifruit juice were recovered in the clarified fraction of the UF process.

Fish and Poultry Processing and Gelatin Industry

In fish processing industry, ultrafiltration is mainly used for fractionation and waste recovery processes. Chabeaud et al. (2009) used UF membrane to improve the bioactivity of a saithe protein hydrolysate containing peptides having a size lower than 7 kDa by fractionating or concentrating some specific molecular weight peptide classes. The wastewaters generated in fish and poultry processing industries contain a large amount of organic load. These wastewaters are usually discharged into the sea without any treatments. The discovery of potentially valuable proteins in the wastewaters in recent years has drawn much attention from several researchers to recover the proteins by membrane filtration. Concentration process was carried out using a ceramic tubular UF membrane (Carbosep M2, MWCO = 15 kDa) and the result showed that UF reduces the organic load from the fish meal wastewaters and allows the recovery of valuable raw materials comprising proteins (Afonso and Bórquez 2002). Afonso et al. (2004) assessed the technical and economical feasibility of protein recovery from fish meal

effluents using cross-flow membrane ultrafiltration and nanofiltration. They concluded that the integrated process comprising MF pre-treatment and UF would enable 69% recovery of proteins allowing for productivity and revenue rise besides a significant reduction of environmental burdens. Therefore, application of UF in fish meal effluents is technically and economically feasible for protein recovery and pollution reduction. On the other hand (Lo et al. 2005) investigated the feasibility of recovering protein from poultry processing wastewater using UF and the optimization of processing parameters. The result pointed out that almost all crude proteins in poultry processing wastewater were retained, subsequently reducing the chemical oxygen demand in the effluent to less than 200 mg L^{-1} .

Membrane Fouling in Food Industry

Fouling refers to the irreversible alteration in membrane properties, resulting from several interactions of feed stream components and membrane (Sablani et al. 2001; Saxena, et al. 2009). In food application, membrane is usually fouled by biofoulants such as protein and polysaccharide (Tsagaraki & Lazarides, 2011). Many authors have studied and proposed the mechanisms of membrane fouling by protein suspensions, which can be grouped as follows:

- The phenomenon of concentration polarization followed by the formation of a gel layer (Blatt et al. 1970; Clifton et al. 1984; Porter 1972)
- Adsorption of solutes on the membrane surface and inside the pore structure (Aimar et al. 1986)
- Deposition and pore blocking of protein aggregates due to denaturation (Martine et al. 1991)

All of these lead to the blockage of the membrane and thereby reduces its flux (Cheryan 1998). Generally, three separate phases of flux decline can be identified as shown in Fig. 1. For example, ultrafiltration of gelatin in a dead-end cell results in a drop in flux to 5% of its initial value in the first minutes (Lim and Mohammad 2010). When macromolecules are filtered and being rejected by the membrane, the molecules that are being rejected will accumulate at the membrane surface, a phenomenon known as concentration polarization. This will subsequently lead to the formation of a gel layer on the membrane surface.

In the second phase, the flux continues to decline but it is due to deposit formation. It is likely that deposition is initially monolayer adsorption and a complete surface layer builds up. In the third phase, a quasi-steady-state period, the flux settles to a steady-state value, which may be due to further deposition of particles or to consolidation of the fouling layer (Marshall et al., 1993). The fact is that, in addition to the decline in flux, the retention of protein

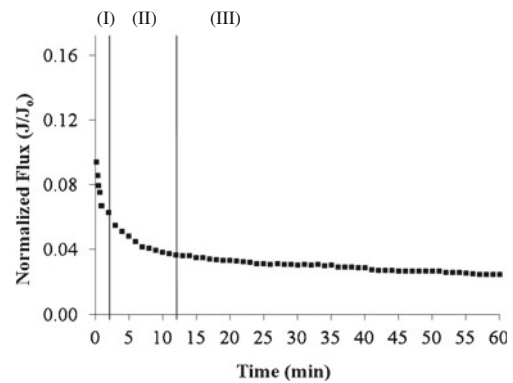


Fig. 1 Three stages of flux decline: *I*, initial rapid drop; *II*, longer-term decline; and *III*, quasi-steady state period. (Lim and Mohammad 2010)

generally increases with time; this is an advantage in UF applications where high protein retention is required.

Membrane fouling, on the other hand, is more complicated in that it is considered as a group of physical, chemical, and biological effects leading to irreversible loss of membrane permeability (Sablani, et al. 2001). Concentration polarization effect usually takes place in less than a minute, whereas fouling takes place over the length of the processing period (Aimar et al. 1991; Nigam, et al. 2008). Fouling and concentration polarization effects are characterized by the state of molecules or solute species involved and by the time scale. Besides, hydrodynamic forces exerted by the flowing fluid and process parameters such as cross-flow velocity, trans-membrane pressure (TMP), feed concentration, pore size and temperature are also factors affecting the rate and extent of membrane fouling and, hence, the permeate flux.

With respect to the membrane characteristics, the hydrophobicity of the top layer is believed to cause the most flux decline (Kimura et al. 2003; Song et al. 2004). For charged organic compounds like protein, electrostatic attraction or repulsion forces between the solute species and the membrane influence the degree of fouling. If the membrane surface charge is not large enough, hydrophobic forces will overcome the electrostatic forces, resulting in more fouling of hydrophobic membranes (Mänttari et al. 2000). Biofouling is another general problem with many membrane processes and involves all biologically active organisms, mainly bacteria and (in some cases) fungi. Biofouling is a dynamic process and involves the formation and growth of a biofilm attached to the membrane. The biofilm may reduce the water flux and even totally prevent water passage (Van der Bruggen et al. 2008).

Membrane fouling is generally associated with cake or gel formation on the membrane surface or blocking membrane pores by macromolecules, colloids, or particulate matters. In situ measurements of fouling and direct observation of cake layer formation are of paramount importance in efforts to understand the fundamental processes governing membrane fouling. Table 1 shows the applications and

Table 1 Different methods of fouled membrane observation

Method	Principle	Application
Direct observation of membrane (Alkhatim et al. 1998)	A microscope objective is positioned at the permeate side of a transparent membrane to observe particle deposition in real time by microscope	To directly observe particle deposition by an optical microscope
Optical laser sensor (Hamachi and Meitton-Peuchot 1999 #82)	The formation of deposit layer absorbs lights from a bypassing laser beam. The variation of the signal intensity after the laser beam traversed through the cake layer corresponds to the deposit layer thickness	To investigate the thickness of cake layer during microfiltration
Ultrasonic time-domain reflectometry (Mairal et al. 2000 #83)	This technique uses sound waves to measure the location of a moving or stationary interface and can provide information on the physical characteristics of the media through which the waves travel	To investigate in situ measurement of membrane fouling Provide information on the physical characteristics of the media
Electrical impedance spectroscopy (Chilcott et al. 2002 #87; Gaedt et al. 2002 #86)	An alternating current is injected directly into the membrane. Capacitance dispersion changes are measured to monitor in situ accumulation of particulates	To characterize membrane properties and to investigate membrane fouling
Scanned electron microscopy	SEM shows 3D images of cake and membrane at much higher magnification	To investigate the membrane surface and fouling

principles of different methods of visual observation of fouling.

Fouling Control

Membrane fouling is an inevitable issue in membrane technology. Techniques to control and minimize the effect and extent of fouling are emerging and developing to ensure that membrane technology is favorable and competitive to other technologies. Management of membrane fouling is an essential topic to investigate to make the successful operation of membrane filtration process. Its avoidance may not be possible, but the impact can be reduced by a variety of techniques. The choice of membrane, module, process configuration, membrane cleaning, and pretreatment are all important in order to reduce membrane fouling. For an installed plant, the options for fouling abatement become more limited. They are more focused on the physical and chemical methods which are summarized in Table 2 (Williams and Wakeman 2000).

Membrane Materials and Modification

Membrane surface modification

Ultrafiltration shares a major portion in protein separation but suffers severe fouling due to the commonly inherent hydrophobic property of membrane surface (Ma et al. 2007). Membrane modification is potentially the most sustainable solution to obtain fouling-resistant membranes (Al-Amoudi and Lovitt 2007). The idea is to insert hydrophilic groups into a polymeric structure so that the overall material becomes more hydrophilic and thus less prone to (organic) fouling. A hydrophobic

polyacrylonitrile (PAN) ultrafiltration membrane was grafted with polyethylene glycol (PEG) to enhance its hydrophilicity and antifouling ability. All prepared polyethylene-graft-polyacrylonitrile (PEG-g-PAN) ultrafiltration membranes showed lower bovine serum albumin (BSA) adsorption, higher flux for protein solution, higher flux recovery ratio, and lower membrane fouling during protein ultrafiltration (Su et al. 2009). Other types of membranes such as polyether sulfone (PES)

Table 2 Methods for reducing flux degradation (Williams and Wakeman 2000)

	Physical	Chemical
Pretreatment	Pre-filtration	Precipitation Coagulation/flocculation Use of disinfectants Use of anti-scalants Adsorption
Design	Use of turbulence promoters Pulsed/ reversed flow Rotating/vibrating membranes Additional fields (e.g., electric)	Choice of membrane material Membrane surface modification
Operation	Limit trans-membrane pressure Maintain a high cross-flow Periodic hydraulic cleaning Periodic mechanical cleaning	Choice of cleaning chemicals Frequency of cleaning

(Taniguchi and Belfort 2004; Rahimpour 2011), polysulfone (PS) (Kaeselev et al. 2001), and polyvinylidene fluoride (PVDF) (Chiang et al. 2009) were also studied for the effects of grafting on membrane performance. Asatekin et al. (2007) reported the use of amphiphilic comb copolymer as an additive in the manufacture of novel PAN UF membranes. Their work showed that the blend membranes prepared with 20 wt.% PAN-g-PEO (combined PEO content, 39 wt.%) were found to resist irreversible fouling by 1,000 ppm solutions of BSA, sodium alginate, and humic acid, recovering the initial pure water flux completely by a pure water rinse or a backwash in the case of humic acid.

Besides surface graft polymerization, various methods such as coating (Hatakeyama et al. 2009; Ju et al. 2009), chemical modification, and photo-modification (Yamagishi et al. 1995) have been presented to reduce UF membrane fouling during protein separation. Su et al. (2008) modified PES membrane with 2-methacryloyloxyethylphosphorylcholine (MPC). The adsorption amounts of BSA on the 2-methacryloyloxyethylphosphorylcholine-modified polyether sulfone (MPC-modified PES) membranes were dramatically decreased in comparison with the control PES membrane. Amphiphilic Pluronic F127 was introduced into PES membranes as both surface modifier and pore-forming agent. The surface hydrophilicity of the PES/Pluronic F127 membranes increased with the increase of Pluronic F127 content and the total fouling and irreversible fouling of the modified membranes remarkably decreased. It was found that these membranes exhibited higher flux recoveries after cleaning (Zhao et al. 2008).

Nanoparticles have also been the focus of numerous studies in recent years to increase the antifouling properties of membrane. Particularly, titanium dioxide (TiO_2) was used to modify PES membrane due to its high photocatalytic and hydrophilicity effects (Razmjou et al. 2011). Different methods of coupling titanium dioxide (TiO_2) on PES membrane were studied and it was found that coating titanium dioxide (TiO_2) on membrane surface is an advanced method compared to entrapping titanium oxide (TiO_2) particles in the membrane matrix for PES membrane modification (Luo et al. 2005; Rahimpour et al. 2008). Studies on the effect of titanium dioxide (TiO_2) nanoparticle size on the performance of PVDF membrane showed that the smaller nanoparticles could improve the antifouling property of the PVDF membrane more remarkably (Cao et al. 2006). Biological fouling can be reduced by the addition of, e.g., silver nanoparticles in the membrane structure (Seung Yun et al. 2007).

Charged Membrane

New membranes with charged characteristics have drawn considerable attention in recent years because of its better fouling resistance. It involves both size- and charge-based

separation processes rather than simply size-based separation process. From this point of view, charged membranes have obviously better separation characteristics (high selectivity and throughput) compared to uncharged membranes. The membrane surface charge can be exploited to improve the selectivity of protein separation processes by adjusting the magnitude of the electrostatic interactions between charged proteins and the charged membrane (Nakao et al. 1988; van Reis et al. 1999). Nakao et al. (1988) proved this through their experimental work on separation of protein mixture (myoglobin and cytochrome C) by charged ultrafiltration membranes. Hydrophilic and charged ultrafiltration membranes through blend PAN and quarternized poly(2-*N,N*-dimethyl aminoethyl methacrylate) were prepared by phase inversion and tested on concentration and purification of collagen (Shen et al. 2009). The separation performance using plate-and-frame modules with charged membranes (cellulose phosphate and diethylaminoethyl cellulose) was investigated for the mixture of BSA, lysozyme, and γ -globulin (Lin and Suen 2002).

Inorganic Membrane

Ultrafiltration membranes are traditionally produced using polymers such as polyethersulfone, polysulfone, cellulose acetate, and regenerated cellulose. However, these polymeric membranes are susceptible to chemical degradation by strong chemical cleaning solutions where membrane life-span is greatly shortened. In addition, some polymeric membranes have limited mechanical stability, leading to a reduction in permeability at high pressures and possible membrane failure in systems employing physical cleaning such as rapid high-pressure backpulsing (Shah et al. 2007). All these drawbacks have motivated the development of a variety of inorganic ultrafiltration membranes with greatly enhanced chemical, thermal, and mechanical stability (Bhave 1991).

Many researches have been conducted to study the performance of ultrafiltration using ceramic membrane. Vladislavljevic et al. (2003) used ceramic tubular UF membranes composed of thin permeate-selective skin of zirconium oxide and titanium dioxide supported by a porous carbon substructure with different molecular weight cutoffs (300,000, 50,000, and 10,000 Da) to clarify depectinized apple juice. A decline in permeate flux over time was observed due to the formation of a layer of retained juice solids on the surface of the membrane that increased overall hydraulic resistance. The fouling resistance decreased with feed flow rate at a transmembrane pressure below 300 kPa. Erdem et al. (2006) prepared ceramic membrane by dip-coating membrane support (alumina) with zirconia sol. The prepared membrane has good protein and lactose separation properties with relatively high protein content (PR~80%) and with

relatively low lactose retention (LR~7%). The permeate flux value was relatively high at around 40 l/m² h.

Electro-ultrafiltration

The application of an electric field to improve the efficiency of pressure-driven filtration processes has been practiced for quite a long time. Electro-ultrafiltration (EUF) is an effective method to decrease gel layer formation on the membrane surface and to increase the filtration flux, primarily due to electrophoresis. Electro-osmosis was found to be significant in some cases when an electric field was applied across the membrane (Joseph et al. 1977; Radovich and Behnam 1983). The basic principle of EUF is related to force balance of charged particle as illustrated in Fig. 2. This force balance includes the 301 forces in the permeate flow direction. If the drag force of the permeate exceeds the oppositely directed lift force, a deposition of the particle will occur (Weigert et al. 1999). The applied electric field drives the charged molecules away from the membrane surface and thus reduces concentration polarization layer (Saxena, et al. 2009). A threefold flux increase was reported by Oussedik et al. (2000) when filtering BSA solutions. The filtration performance during EUF has been tested with several industrial enzyme solutions. Results showed that EUF is an effective method to filter highly concentrated solutions at low cross-flow. The flux improved three to seven times for enzymes with a significant surface charge at electrical field strength of 1,600 V/m compared to conventional UF. The greatest improvement is observed at a high concentration. A three- to seven-time flux increase is obtained compared to conventional cross-flow UF for two amylase solutions (Enevoldsen et al. 2007). Sarkar et al. (2008a) observed 32% enhancement of permeate flux when an external electric field was applied during clarification of mosambi juice (*Citrus sinensis* (L.) Osbeck) using a flat sheet of polyethersulfone membrane (50 kDa MWCO) in cross-flow ultrafiltration under laminar flow conditions. Instead of a constant field, a pulsed electric field can also be

used. A pulsed electric field consumes less energy than a constant field, and for some systems a pulsed electric field results in an even higher flux compared to a constant field (Oussedik, et al. 2000; Weigert, et al. 1999). A conventional cross-flow ultrafiltration (CUF) apparatus was modified by the inclusion of electrodes which permitted a pulsed electric field to be produced across the ultrafiltration membrane (PEF-UF process). Studies of the process with BSA in the range of 0.5–5% w/v demonstrated 25–40% decrease in solute-related resistance to the permeate flux compared to the case of a zero electric field. Accordingly, higher permeate fluxes and, therefore, higher rates of concentration of the protein solution were obtained than for conventional cross-flow ultrafiltration. When the electric field was reimposed following a period of operation under conventional CUF conditions, the permeate flux could be restored to nearly the same value observed initially for the PEF-UF process (Robinson et al. 1993). Sarkar et al. (2008b) studied the effects of pulsed electric field during cross-flow ultrafiltration of synthetic juice (mixture of sucrose and pectin). It was observed that, with an increase in electric field and pulse ratio, permeate flux increases.

Ultrasonic Field

Ultrasound has gained increasing attention as a technique of fouling control in recent years. Several different mechanisms may lead to particle release from a particle-fouled surface as a result of ultrasound. The proposed mechanisms illustrated in Fig. 3 include acoustic streaming, microstreaming, micro-jet, and microstreamers (Lamminen et al. 2004). Acoustic streaming is defined as the absorption of acoustic energy resulting in fluid flow, whereas microstreaming is a time-independent circulation of fluid occurring in the vicinity of bubbles set into motion by oscillating sound pressure. Oscillations in bubble size cause rapid fluctuations in the magnitude and direction of fluid movement, and as a result significant shear forces occur. Cavitation bubbles that form at nucleation sites within the liquid and are subsequently translated to a mutual location (antinodes) are called microstreamers. Micro-jets are formed when a cavitation bubble collapses in the presence of an asymmetry (i.e., a surface or another bubble). During collapse, the bubble wall accelerates more on the side opposite to a solid surface, resulting in the formation of a strong jet of water.

Acoustic streaming does not require the collapse of cavitation bubbles. It is expected to be important near surfaces with loosely attached particles or with readily dissolvable surfaces (Lamminen et al. 2004). Higher-frequency ultrasound tends to have higher energy absorption and thus greater acoustic streaming flow rates than lower frequencies for the same power intensity (Suslick 1988). This mechanism causes bulk water movement toward and away from

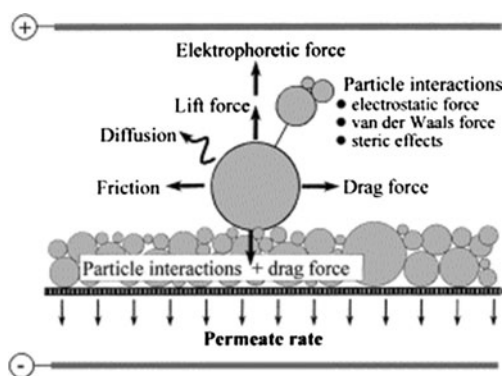


Fig. 2 Force balance on a particle during the filtration process (Weigert et al. 1999)

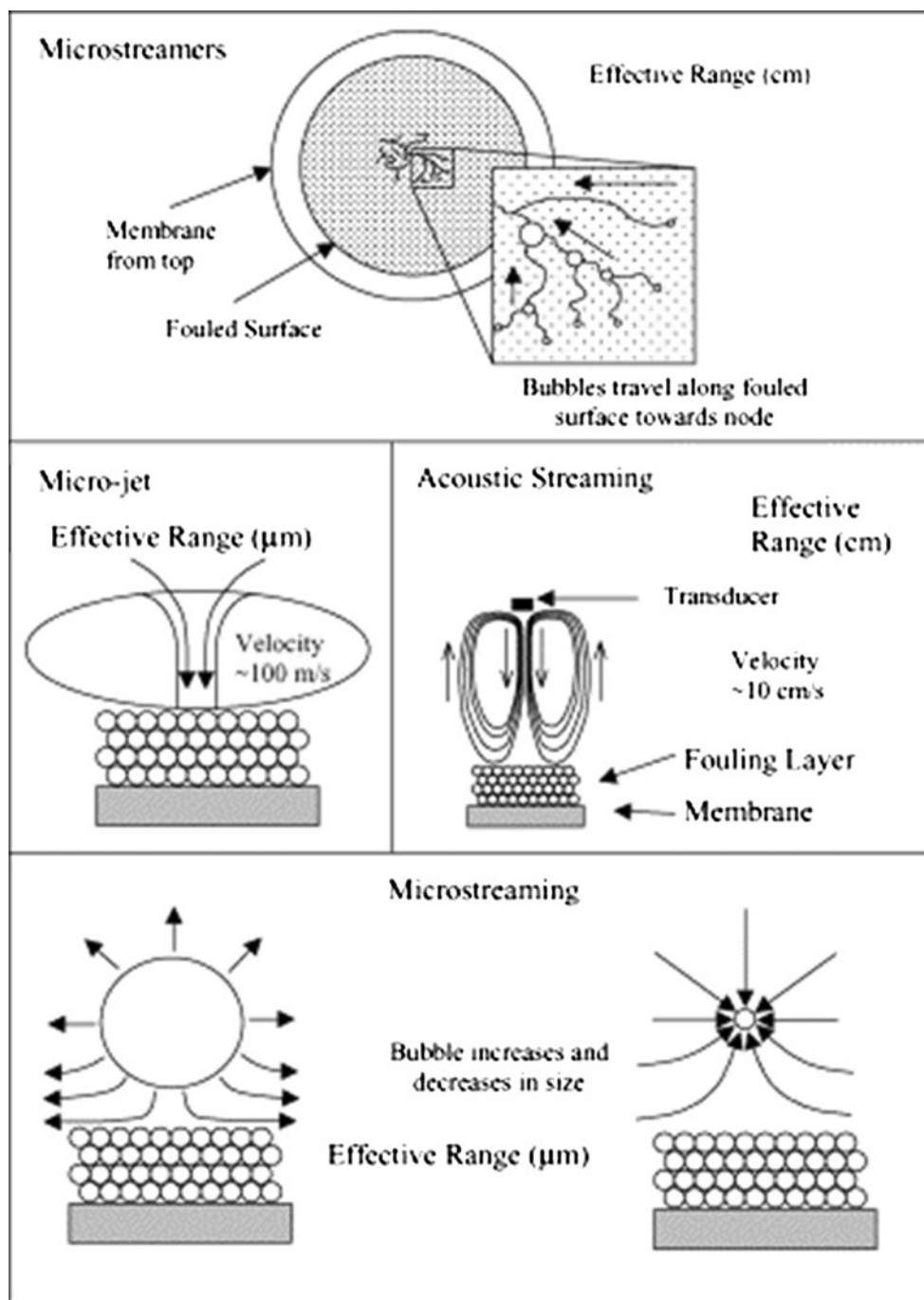


Fig. 3 Possible mechanisms for particle removal/detachment observed with ultrasonic cleaning (Lamminen et al. 2004)

the membrane cake layer, with velocity gradients near the cake layer that may scour particles from the surface. While microstreaming, microstreamers, and micro-jets are caused by cavitation bubbles, they are also able to scour particles from a membrane surface to different extents, respectively (Lamminen, et al. 2004).

There are not much works done on the effect of ultrasonic field on separation process in food industry. Muthukumaran

et al. (2005) observed that ultrasonic radiation at low power levels can significantly enhance the permeate flux with an enhancement factor of between 1.2 and 1.7. Furthermore, the use of turbulence promoters (spacers) in combination with ultrasound can lead to a doubling in the permeate flux. The concentration profile of the whey proteins before and after sonication was also not affected by the sonication process for both cases. They extended their study and found

out that the use of continuous low frequency (50 kHz) ultrasound was most effective in both the fouling and cleaning cycles whereas the use of intermittent ultrasound did little to enhance flux rates at any frequency. There were conditions under which it could even have a negative effect on filtration performance. For instance, the use of intermittent (pulsed) ultrasound at high frequency (1 MHz) caused a net reduction in flux rates when high transmembrane pressures and low cross-flow velocities were employed (Muthukumaran et al. 2007).

The effect of ultrasound on the flux and solute rejection in cross-flow UF of binary BSA and lysozyme (Ly) using PS membrane (MWCO, 30,000) has been studied. Ultrasonic irradiation not only enhanced the UF flux but also increased the Ly rejection to some extent. The use of ultrasound at 25 kHz and 240 W resulted in an increase of UF flux by 135% and 120% with PS membrane at pH 11 in the upward and downward modes, respectively, in contrast to the case without ultrasound (Teng et al. 2006). Iritani et al. (1997) reported that ultrasonic irradiation contributed to the remarkable improvement in the filtration rate and lysozyme rejection in upward ultrafiltration of binary BSA/lysozyme mixtures.

Masselin et al. (2001) studied the effect of 47 kHz ultrasonic waves on PES, PVDF, and PAN membranes and reported that only PES is affected by the ultrasonic treatment over its entire surface. PVDF and PAN membranes are more resistant and present less damages at the exception of the PAN50 and the PVDF40 membranes for which the edges are more affected than the central section. Results also show that the degradation of the membrane surfaces under ultrasonic stress leads to an increase in pore radius for large pores, an overall increase in pore density and porosity, and the formation of large cracks preferentially at the edges of the membrane samples. Muthukumaran et al. (2005) also showed that ultrasonic radiation did not alter PS membrane integrity. From these findings, it can be concluded that, in spite of their great efficiency in enhancing permeation of fouled membranes, ultrasounds have to be used with care. The nature of the polymeric material as well as the ultrasonic wave frequency and intensity have to be taken into account (Masselin, et al. 2001).

Hydrodynamic Methods

Flow Manipulation

Although membrane fouling can be reduced by modifications to the properties of the feed and the use of new or modified membranes and external force fields, it cannot be eliminated completely. Flow manipulation by controlling hydrodynamics such as transmembrane pressure and permeate flux is another

important strategy to combat both reversible and irreversible fouling of membrane (Gésan et al. 1993).

An intrinsic solution to the problem of membrane fouling could be the concept of critical flux. Critical flux is the maximal flux where fouling remains reversible; when operating below the critical flux, flux decline can be reversed by non-destructive measures. The critical flux concept represents the shift from repulsive interaction (dispersed matter–polarized layer) to attractive interaction (condensed matter–deposit). Several researchers have showed that critical flux may increase with enhancing cross-flow rate (which also could be expressed as Reynolds number or shear stress), decreasing feed concentration, and also increasing membrane pore size (Chiu et al. 2006; Mänttari and Nyström 2000; Metsämuuronen et al. 2002; Youravong et al. 2003). There is another concept evolved from the critical flux theory and which can be considered a generalization. This concept is known as sustainable flux. It is defined as the flux above which the rate of fouling is economically and environmentally unsustainable. The sustainable flux depends on hydrodynamics, feed conditions, and process time and is therefore hard to determine (Bacchin et al. 2006). Nevertheless, understanding of this principle leads to guidelines for operational conditions where fouling is minimized (Nyström et al. 2003; Stoller and Chianese 2006).

Work had been done to determine the critical flux of skimmed milk and to investigate the effects of hydrodynamics and protein concentration on the critical flux for two different membranes. The critical flux decreased as the protein concentration increased and increased as the wall shear stress increased (Youravong, et al. 2003). In an operation of milk ultrafiltration, the fluid flux through the membrane initially increased with the increase in operating pressure. Any further increase in operating pressure brought about no change in flux. The effect of operating temperature was also investigated. Experiments were conducted at different temperatures (from room temperature up to 50 °C), at a constant transmembrane pressure of 200 kPa. As temperature increases, flux is improved, which reflects the expected influence of temperature on viscosity and mass transfer coefficient, which improves the transfer of milk components from the membrane surface back into the bulk stream (Makardij et al. 1999). Vladislavljevic et al. (2003) studied the effect of operating parameters such as transmembrane pressure, feed flow rate through the module, and temperature on the permeate flux and fouling resistance in apple juice UF. The steady-state fouling resistance increased with transmembrane pressure and at 400 kPa reached more than 93% of the total resistance. For small transmembrane pressures, e.g., 100 kPa, the fouling resistance significantly decreased with increasing feed flow rate, which was due to a higher rate of solute back-transfer. At relatively high operating pressures (above 300 kPa), the steady-state fouling resistance,

Table 3 Examples of cleaning solutions and their applications (Williams and Wakeman 2000)

Type of cleaning solution	Effectivity against typical foulants
Mineral acids, sodium hexametaphosphate polyarylates, ethylenediaminetetra-acetic acid	Salt precipitates, mineral scalants
Sodium hydroxide-based cleaners, with or without hypochlorite	Solubilising fats, proteins
Enzyme cleaner based on proteases, amylases, and glucanases	Used in specific instances at neutral pH

i.e., permeate flux, was virtually independent on the feed flow rate. Under these conditions, permeate flux was limited by the dense structure of the deposited fouling layer.

Another interesting technology that has been introduced in the last few years is based on the vibrating shear-enhanced membrane filtration system (Jaffrin 2008; Akoum et al 2004). Such a system uses oscillatory vibration to create high shear at the surface of the filter membrane. This high shear force significantly improves the membrane's resistance to fouling, thereby enabling high throughputs and minimizing reject volumes (Kertész et al 2010). Several studies have been carried out using the vibrating membrane system such as for concentration of milk (Akoum et al 2005) and separation of enzymes and yeast cell (Beier and Jonsson 2007).

Turbulence Promoters

The flow field generated by a static mixer induces hydraulic turbulence and increases the wall shear stress in the membrane, which leads to enhanced scouring of the membrane surface and therefore to the permeate flux enhancement. This technique has limited application because it can easily damage the integrity of a membrane and hence reduce its lifespan. However, recent development of ceramic membranes induced a moderate revival in the use of static turbulence promoters in cross-flow membrane filtration. Therefore, the use of static turbulence promoters as a method for reducing concentration polarization and membrane fouling in cross-flow membrane filtration has been investigated relatively often (Krstic et al. 2006). Bellhouse et al. (2001) studied the detailed fluid dynamic processes contributing to flux enhancement when screw thread inserts are used with tubular membranes. Filtration tests under typical micro-filtration, ultrafiltration, and nanofiltration conditions all showed dramatic increases in filtration fluxes (by factors of 6–10) when membrane systems with inserts were compared with plain tubular membranes at the same cross-flow rate. However, the inserts cause higher pressure drops than the plain membranes under the same operating conditions. Another relevant work investigated the permeate flux and the specific energy consumption during ultrafiltration of the endo-pectinase solution obtained during the operation without and with the static mixer. The flux

enhancement of 45% with the reduction of the specific energy consumption of 40% was achieved when the static mixer was used compared to the operation without the static mixer (Krstic et al. 2007).

Backwashing and Backpulsing

Backpulsing and other comparable techniques, such as backwashing, backflushing, and backshocking, are also effective alternatives to remove fouling (Redkar et al. 1996). In these procedures, the transmembrane pressure is inverted and part of the permeate flows backward into the cross-flow channel. Backwash pressures need to be greater than the operating filtration pressure. This technique is limited to removal of surface deposits from the membrane. It may be ineffective when the deposits adhere strongly or if membrane pores were fouled (Wakeman and Williams 2002). The effectiveness of this technique depends mainly on the pulse frequency and duration. Besides, it is also highly dependent on the feed composition and the pressure profile.

Gas Sparging

Gas sparging is a method proposed for reducing concentration polarization and membrane fouling by injecting air into the feed stream, creating a gas–liquid two-phase flow across the membrane surface. The injected air promotes turbulence, increasing the superficial cross-flow velocity of the process fluid, suppressing the polarization layer, and enhancing the ultrafiltration process (Cui and Wright 1994). Many studies on the effect of gas sparging on flux enhancement in various ultrafiltration processes had been carried out extensively (Bellara et al. 1996; Cui and Wright 1996; Ducom and Cabassud 2003; Ghosh 2006; Li et al. 1998; Li et al. 2008; Sur and Cui 2005). Cui and Wright (1994) observed an increase of up to 250% in permeate flux in air-sparged ultrafiltration of dextrans and BSA using vertical tubular membranes. The combined impact of cross-flow rate and gas sparging on critical flux, limiting flux, and selectivity was studied by a total recycle mode using a hollow fiber membrane with molecular weight cutoff of 30 kDa (Li, et al. 2008). Nevertheless, gas sparging gave a negative effect on soluble protein and peptide

Table 4 Summary of reported works on fouling in food industry and expected future work

Aspects	Main findings	Main references	Future work
Fouling mechanism and identification of factors influencing the fouling	Early works focused on this issue. The mechanisms and factors influencing fouling are now quite established in many cases. Biofouling has recently been the focus as well	Sablani et al. (2001) Song et al. (2004) Mänttari et al. (2000) Van der Bruggen et al. (2008)	Fouling due to the presence of complex solution such as that found in food industry has not been well understood. There are areas that still can be explored with regard to mechanism of fouling, modeling, and biofouling
Fouling control			
Development of new membrane types, surface modification, charged membranes, inorganic membranes	Insertion of hydrophilic groups into a polymeric structure so that the overall material becomes more hydrophilic and thus less prone to (organic) fouling has been the main focus in recent studies	Rahimpour (2011) Asatekin et al. (2007) Hatakeyama et al. (2009) Ju et al. (2009) Shen et al. (2009) Erdem et al. (2006)	The role of nanotechnology in the surface modification of membranes is increasingly being explored. New materials (polymeric and inorganic) are still being developed by various researchers
Electroultrafiltration	It is an effective method to decrease gel layer formation on the membrane surface and to increase the filtration flux, primarily due to electrophoresis. However, application in a large scale is still hindered due to cost	Enevoldsen et al. (2007) Sarkar et al. (2008a)	The focus should be on scaling up the process that is cost-effective and practical
Ultrasonic	Interesting findings at lab scale but the issue is still in application in a large scale due to cost-effectiveness factor	Masselin et al. (2001) Muthukumaran et al. (2005)	The focus should be on scaling-up the process that is cost-effective and practical
Hydrodynamic methods, flow manipulation, turbulence promoters, backwashing and backpulsing, gas sparging	Well established and has been in practice	Krstic et al. (2007) Vladisavljevic et al. (2003) Wakeman and Williams (2002) Cui and Wright (1994)	The manipulation of hydrodynamic in designing new membrane modules. This can be achieved through computational fluid dynamic modeling and utilizing the findings for a new improved design
Membrane cleaning	Chemical cleaning is quite well established. Proprietary cleaning solutions are available and being used in practice. The choice of cleaning solution is not only determined by the foulant type but also by the compatibility of the membrane with the solution at the cleaning temperature. Enzymatic cleaning has been a major focus recently	Al-Amoudi and Lovitt (2007) Kazemimoghdam and Mohammadi (2007) Petrus et al. (2008) Argüello et al. (2003)	New chemicals and enzymes that will provide milder cleaning and less frequent regimes should be investigated. Again, nanotechnology approach can play an important role to obtain new cleaning agents

transmission and resulted in the decay of selectivity at subcritical condition and critical flux condition. There is also problem in handling the gas injected into the membrane system and getting the desired air bubble size. In addition, gas sparging could also cause unwanted foaming of milk in the module and denaturation of protein (Brans et al. 2004).

Membrane Cleaning

Nevertheless, over long periods of operation, membrane fouling is generally not totally reversible by the hydraulic backwash procedure. As the number of filtration cycle increases, the irreversible fraction of membrane fouling also increases. In order to obtain the desired production flow rates or flux, an increase in TMP is required. When this pressure reaches the maximum allowed by the mechanical resistance of the membrane, chemical cleaning of the membrane is required for the membrane to restore most of its initial permeability (Crozes et al. 1997).

Fouled membranes are commonly rejuvenated by using cleaning in place (CIP) procedures. CIP can improve performance with shorter downtimes than cleaning out of place. Cleaning solutions are usually circulated with a pressure somewhat lower than that used during filtration to prevent deeper penetration of the foulants into the membrane. Proprietary cleaning solutions are available. Some general information about types of cleaning solutions are given in Table 3 (Williams and Wakeman 2000). The choice of cleaning solution is determined not only by the foulant type but also by the compatibility of the membrane with the solution at the cleaning temperature.

There are many types of cleaning agent available for membrane cleaning and they are categorized as acids, alkalis, surfactants, disinfectants, enzymatic, and combined cleaning materials. Caustic is typically used to clean organic and microbial fouled membranes by hydrolysis or/and solubilization. Oxidants clean membrane by reducing the adhesion of fouling materials to membranes. Surfactants form micelles with fat, oil, and proteins in water and help to clean the membranes fouled by these materials. In addition, surfactants can disrupt functions of bacteria cell walls and hence remove biofilms (Al-Amoudi and Lovitt 2007). Cleaning efficiency is mainly dependent on several factors such as pH, concentration, cleaning time and frequency, and operating conditions (Al-Amoudi and Lovitt 2007; Kazemimoghadam and Mohammadi 2007; Makardij, et al. 1999). Suitable hydrodynamic conditions tend to facilitate mass transfer and thus enhance the efficiency of cleaning. Temperature is believed to have a significant impact on both the efficiency and rate of membrane cleaning by changing the reaction equilibrium, by enhancing the reaction kinetics, and by increasing the

solubility of solutes. Although chemical cleaning can be very effective to remove fouling as compared to other techniques, it severely damages the membrane materials and thus reduces membrane lifespan. To overcome this, various works have been reported on the use of enzymatic cleaners because of their capacity to develop their activity in mild conditions, which is a determining factor for their application in the cleaning of membranes that are sensitive to chemicals, pH, and/or temperature (Petrus et al 2008; Argüello et al. 2003).

Cleaning of fouled membranes is important not only due to economical concern in ultrafiltration but also because there are concerns with regard to environmental problems as cleaning usually discharges chemical waste. Therefore, properly designed and optimized cleaning procedures should be implemented and continuous research and development is essential to counter this issue.

Conclusions

Membrane filtration processes are gaining more attention and focus in food industry due to its advantages (environmental friendliness, cost saving, and product improvement) as compared with other conventional methods. However, membrane can be easily fouled by various solutes, for instance, protein and polysaccharide in food industry. Fouling decreases permeate flux severely and thus increases filtration processing time, which is not economically effective. Therefore, development of fouling control and minimization is crucial to enable membrane technology to play an indispensable role in food industry and others as well. Earlier works have focused on studying the fouling mechanisms and phenomena, while recent studies have focused on membrane fouling control through membrane modifications and use of innovative methods to reduce fouling. Various studies have been reviewed in this paper and summarized as shown in Table 4, including the prospect for further research work.

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